Approximation of areas

1

a 117 b 6 c 2

2

a $\frac{1}{2}$ b $\frac{5}{16}$

3

a 60 b $17\,000$ c 34

4

a 2.426 b 1.167

5

a 180.828 b -21.000 c 0.919 d 0.502

e 1.165

6 5.2

7 1.5

8 256 units^2

9 $\frac{77}{60} \text{ units}^2$

10

a 62 units^2 b 50 units^2

11 16.1

12

a $y'' = 4x^2e^{x^2} + 2e^{x^2}$

b 30.5

c It is an overestimate as e^{x^2} is concave up between $x = 0$ and $x = 2$.

13 19.81 units^2

14 28.9

Applications

15

a $232\,484 \text{ m}^2$

b 8183 m^3

16

a $14\,104 \text{ m}^2$

b 479.54 m^3

17 432 m^2

18 $1\,707\,750 \text{ m}^2$

19 26.8 m^2

20 126 m^2

21 3.12 ha

22 201.95 m^2